



wazuh.

Wazuh – Vulnerability Detection

THREAT DETECTION AND RESPONSE

Lab Created By: MUHAMMAD MOIZ UD DIN RAFAY

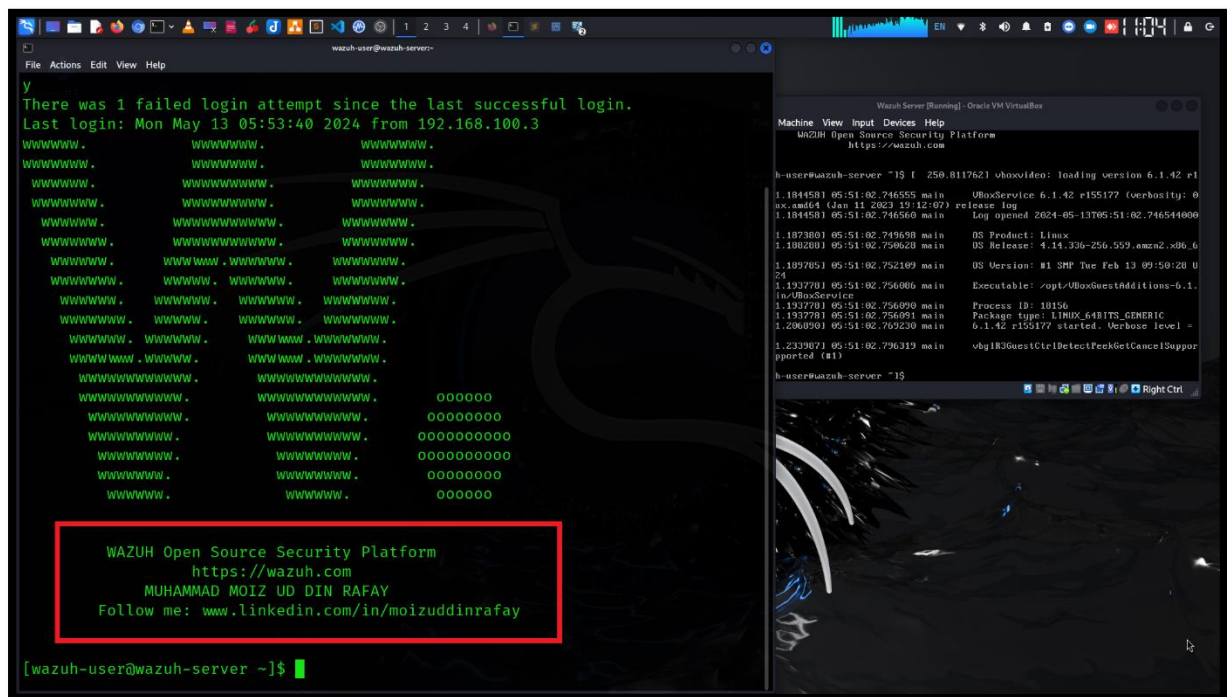
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Wazuh is an open-source security platform that provides threat detection, integrity monitoring, and security analytics. It's designed to help organizations detect and respond to security incidents effectively. One of its key functionalities is vulnerability detection, which plays a crucial role in maintaining the security posture of systems and networks.

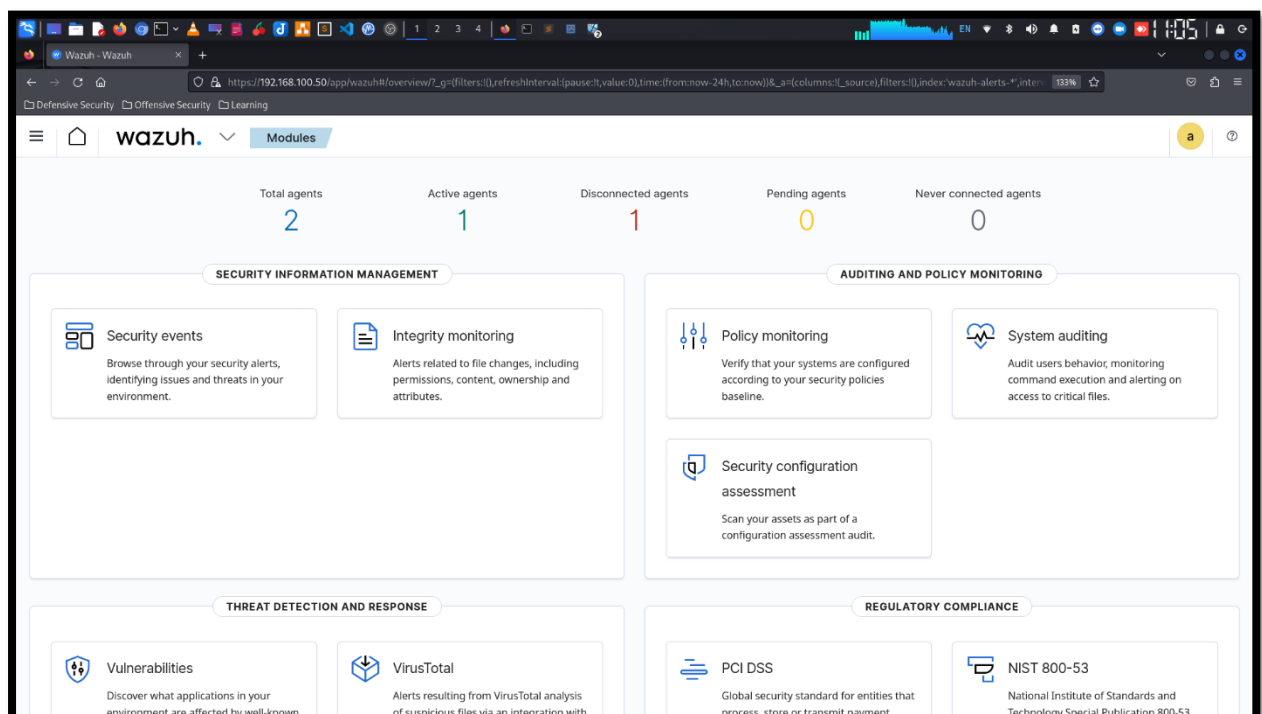
Wazuh employs several techniques for vulnerability detection:

1. **Vulnerability Scanning:** Wazuh integrates with popular vulnerability scanning tools such as OpenVAS and Nessus. These tools scan the network and systems for known vulnerabilities by comparing their configurations and installed software versions against a database of known vulnerabilities.
2. **Asset Inventory:** Wazuh maintains an inventory of assets within the organization's network, including hardware devices, software applications, and their respective versions. By continuously monitoring this inventory, Wazuh can identify discrepancies or outdated software versions that may be vulnerable to known exploits.
3. **Behavioural Analysis:** In addition to scanning for known vulnerabilities, Wazuh monitors system and network behaviour for anomalies that could indicate potential security issues. For example, sudden spikes in network traffic or unexpected changes in system configurations may suggest the presence of an exploit or vulnerability.
4. **Real-time Alerts:** Wazuh generates real-time alerts when vulnerabilities are detected, providing administrators with immediate notification of potential security threats. These alerts include detailed information about the vulnerability, affected systems, and recommended remediation steps.
5. **Customization and Extensibility:** Wazuh is highly customizable and extensible, allowing organizations to tailor vulnerability detection capabilities to their specific needs. Administrators can define custom rules and policies for detecting vulnerabilities based on their unique environment and security requirements.

Here is Wazuh Server running on my lab environment. I access Wazuh console via SSH connection.



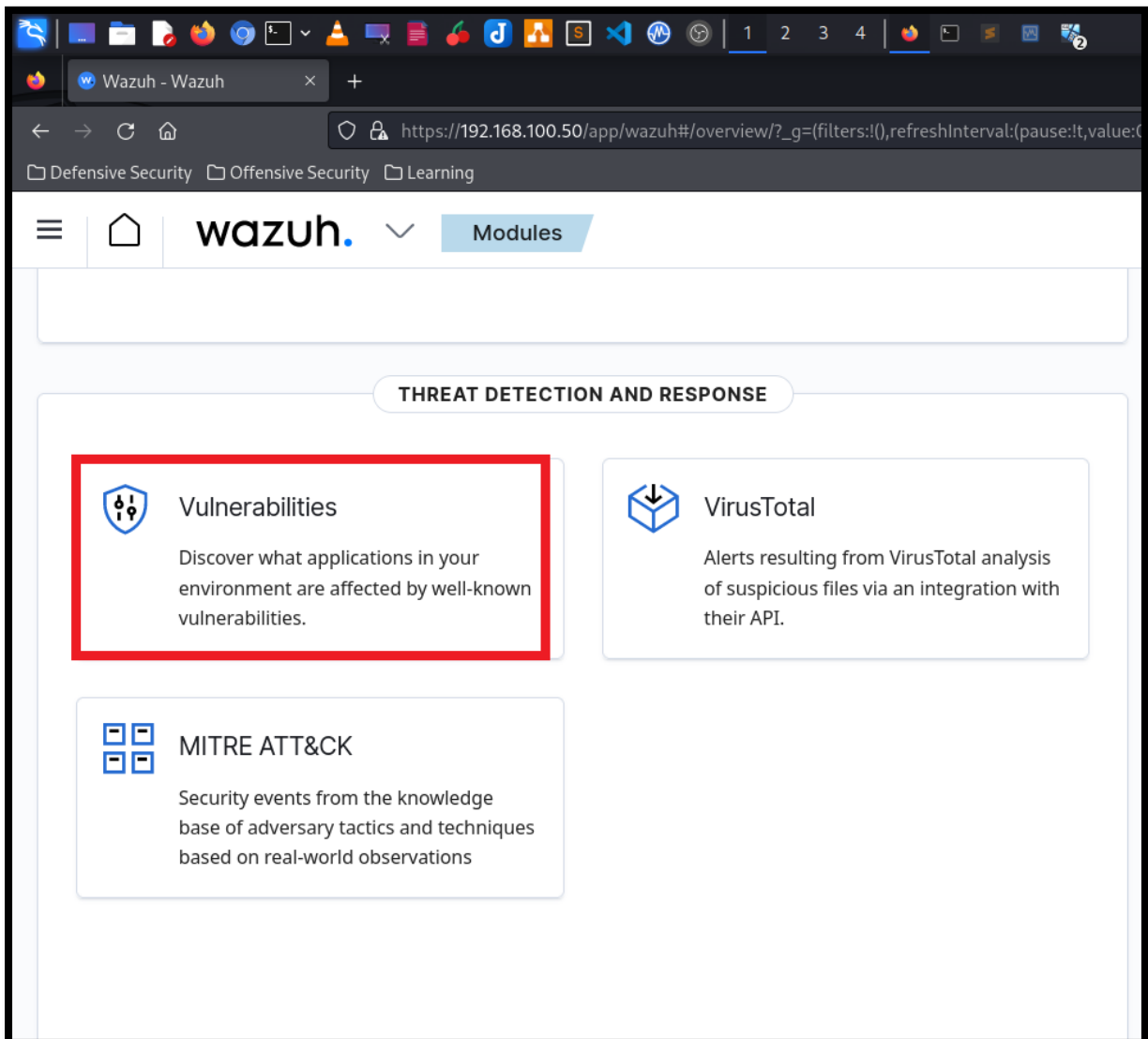
Here is Wazuh Server dashboard.



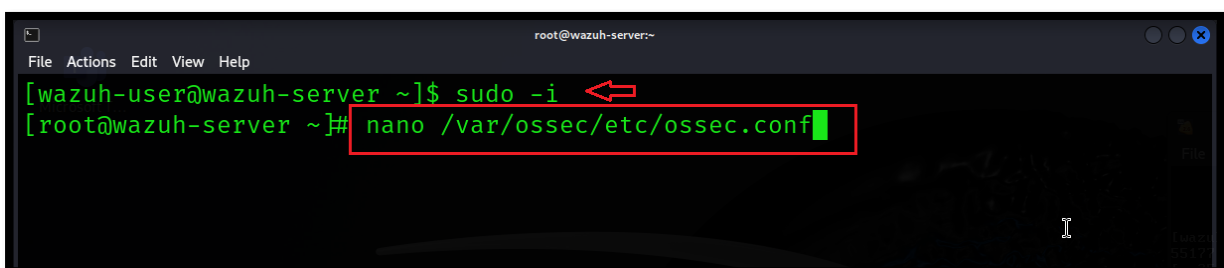
Selecting the Active agent (Windows 11)

Wazuh – Vulnerabilities Detection Lab: 07
Lab Created by: MUHAMMAD MOIZ UD DIN RAFAY

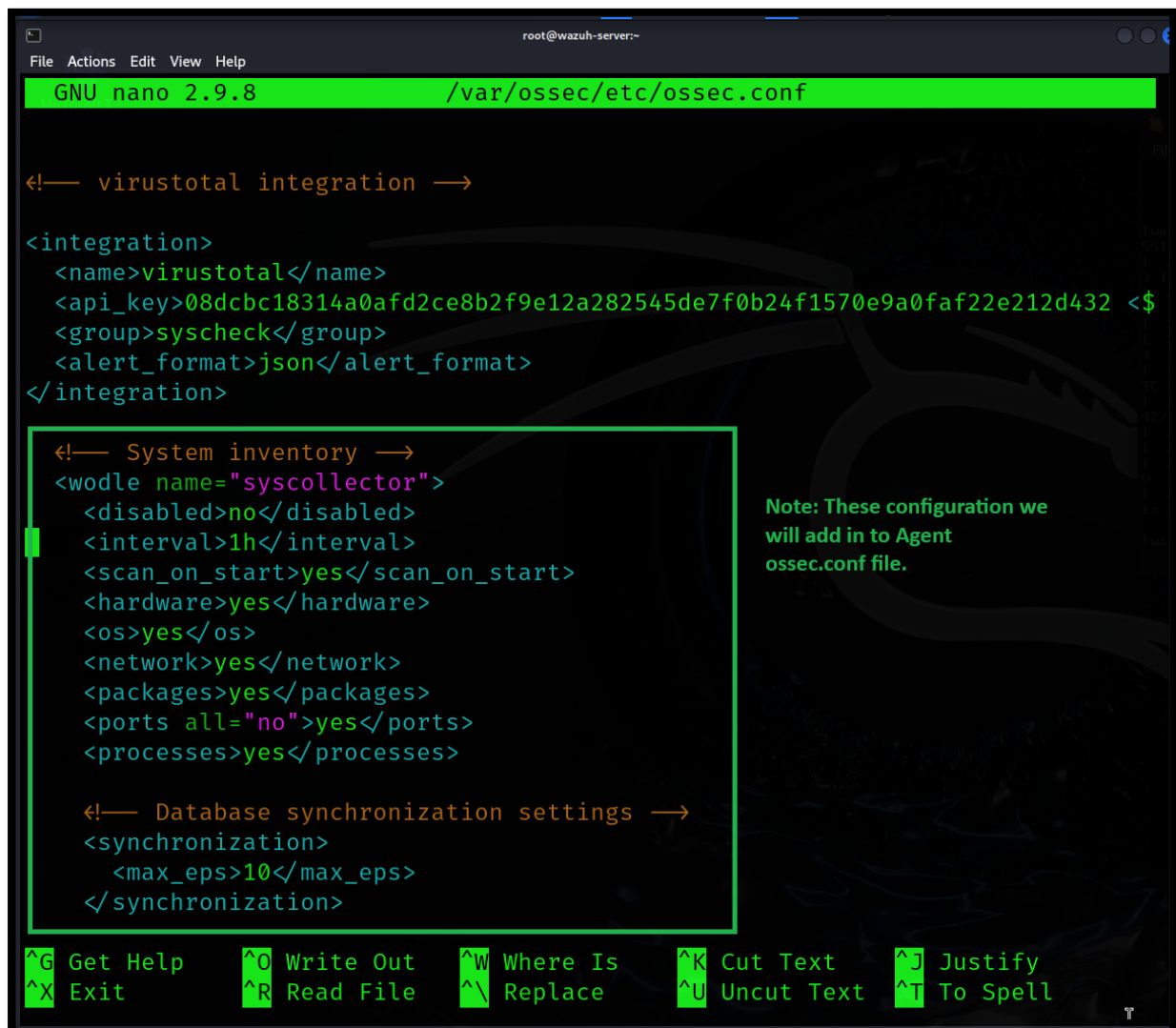
Scroll down and go to “THREAT DETECTION AND RESPONSE” > “Vulnerabilities” options is available.



Now we have to configure vulnerability scanner.
Edit: “nano /var/ossec/etc/ossec.conf” file.



Here is in “ossec.conf” look at “System inventor” this configuration already there, we will add these configuration in to Windows-agent “ossec.conf” later.



```
GNU nano 2.9.8 /var/ossec/etc/ossec.conf

<!-- virustotal integration -->

<integration>
  <name>virustotal</name>
  <api_key>08dcbc18314a0afd2ce8b2f9e12a282545de7f0b24f1570e9a0faf22e212d432 <$
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>

<!-- System inventory -->
<wodle name="syscollector">
  <disabled>no</disabled>
  <interval>1h</interval>
  <scan_on_start>yes</scan_on_start>
  <hardware>yes</hardware>
  <os>yes</os>
  <network>yes</network>
  <packages>yes</packages>
  <ports all="no">yes</ports>
  <processes>yes</processes>

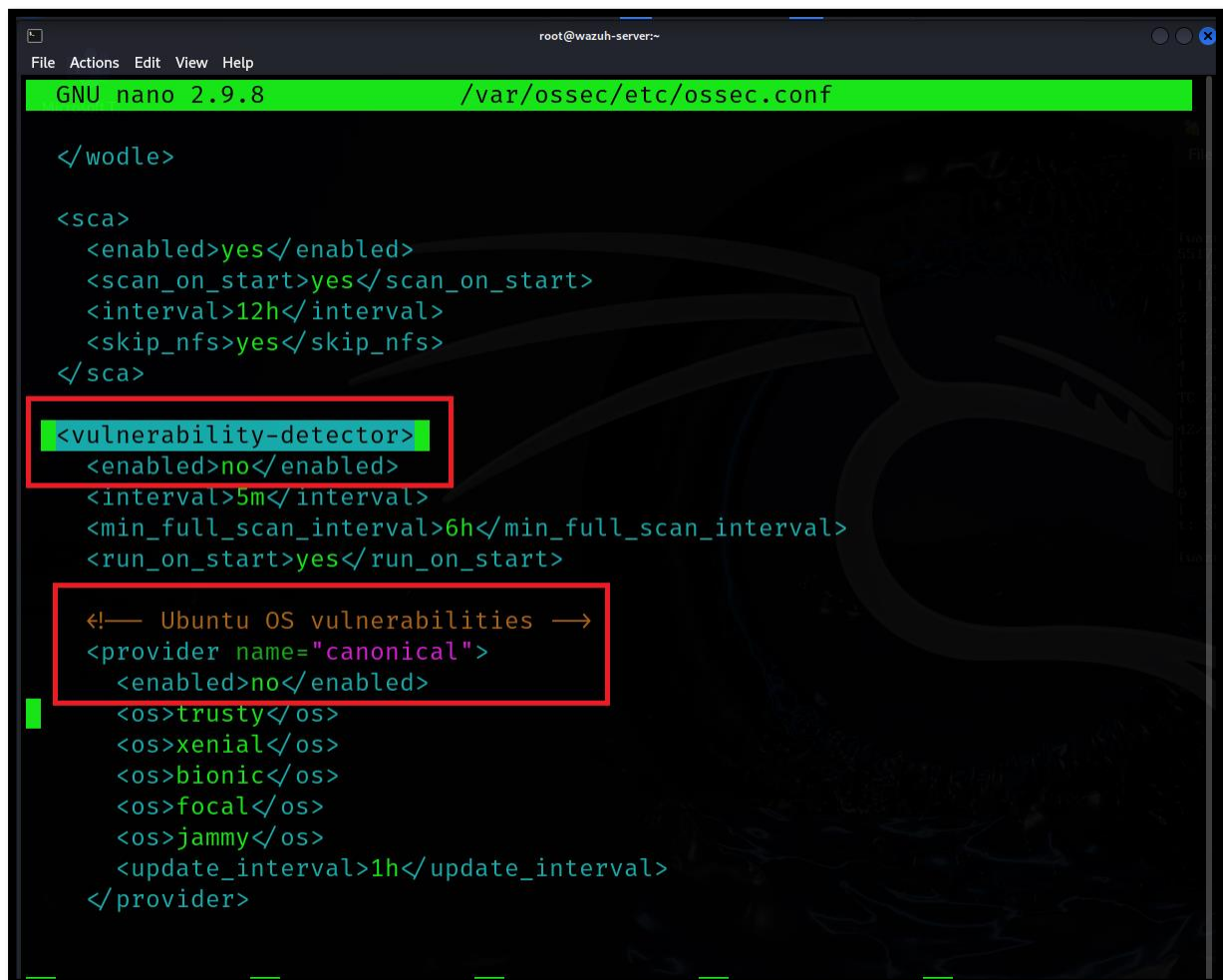
  <!-- Database synchronization settings -->
  <synchronization>
    <max_eps>10</max_eps>
  </synchronization>

^G Get Help      ^O Write Out     ^W Where Is      ^K Cut Text      ^J Justify
^X Exit          ^R Read File     ^_ Replace       ^U Uncut Text    ^T To Spell
```

```
<wodle name="syscollector">
<disabled>no</disabled>
<interval>1h</interval>
<scan_on_start>yes</scan_on_start>
<hardware>yes</hardware>
<os>yes</os>
<network>yes</network>
<packages>yes</packages>
<ports all="no">yes</ports>
<processes>yes</processes>

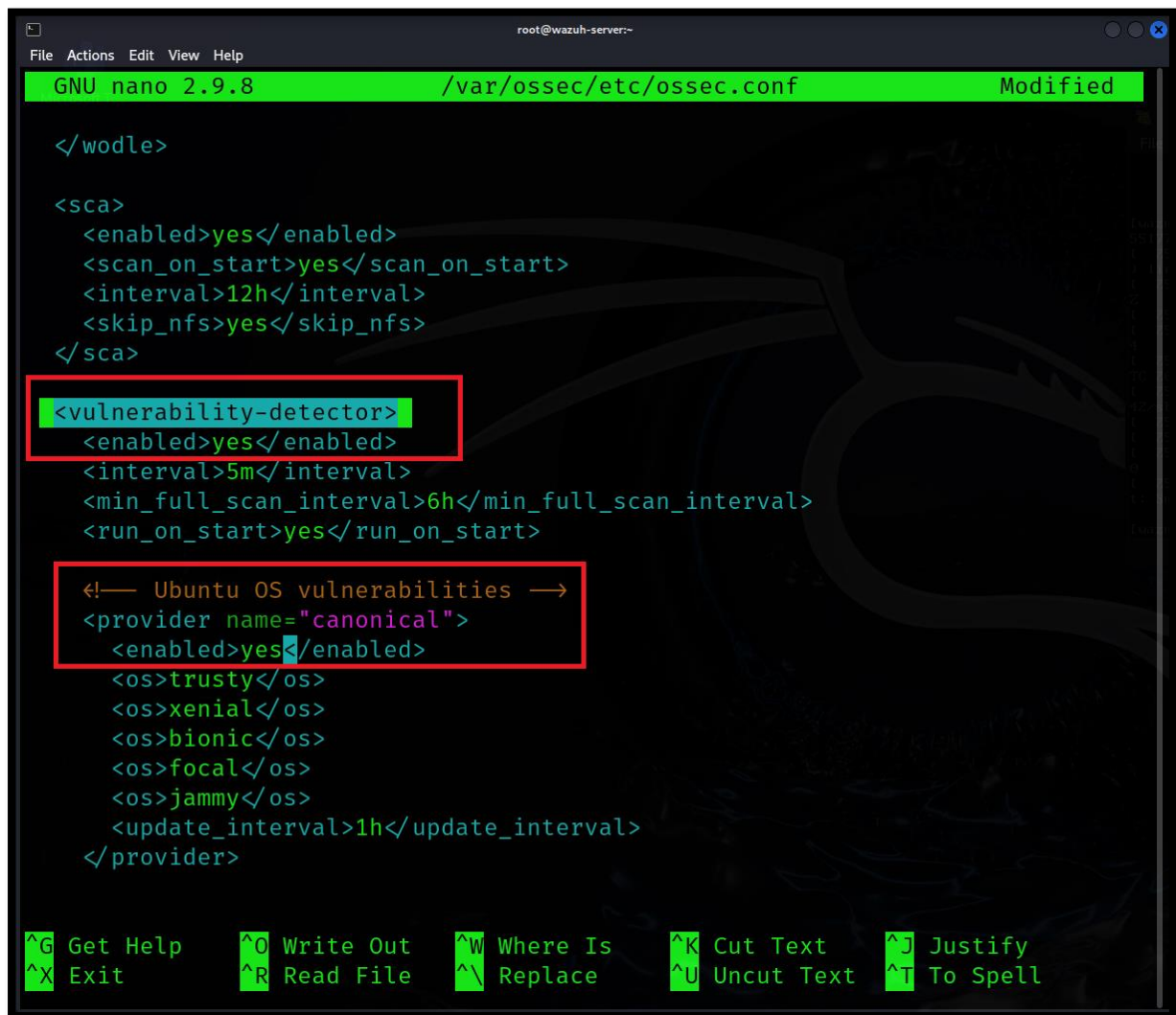
<!-- Database synchronization settings -->
<synchronization>
<max_eps>10</max_eps>
</synchronization>
```

Now scroll down and see there is “vulnerability-detector” not enabled. Also some OS vulnerabilities configuration is set to not enabled.



```
root@wazuh-server:~  
File Actions Edit View Help  
GNU nano 2.9.8 /var/ossec/etc/ossec.conf  
  
</wodle>  
  
<sca>  
  <enabled>yes</enabled>  
  <scan_on_start>yes</scan_on_start>  
  <interval>12h</interval>  
  <skip_nfs>yes</skip_nfs>  
</sca>  
  
<vulnerability-detector>  
  <enabled>no</enabled>  
  <interval>5m</interval>  
  <min_full_scan_interval>6h</min_full_scan_interval>  
  <run_on_start>yes</run_on_start>  
  
  <!-- Ubuntu OS vulnerabilities -->  
  <provider name="canonical">  
    <enabled>no</enabled>  
    <os>trusty</os>  
    <os>xenial</os>  
    <os>bionic</os>  
    <os>focal</os>  
    <os>jammy</os>  
    <update_interval>1h</update_interval>  
  </provider>
```

We have to enable configuration here follow as shown in figure.



```
GNU nano 2.9.8 /var/ossec/etc/ossec.conf Modified

</wodle>

<sca>
  <enabled>yes</enabled>
  <scan_on_start>yes</scan_on_start>
  <interval>12h</interval>
  <skip_nfs>yes</skip_nfs>
</sca>

<vulnerability-detector>
  <enabled>yes</enabled>
  <interval>5m</interval>
  <min_full_scan_interval>6h</min_full_scan_interval>
  <run_on_start>yes</run_on_start>

  <!-- Ubuntu OS vulnerabilities -->
  <provider name="canonical">
    <enabled>yes</enabled>
    <os>trusty</os>
    <os>xenial</os>
    <os>bionic</os>
    <os>focal</os>
    <os>jammy</os>
    <update_interval>1h</update_interval>
  </provider>

^G Get Help      ^O Write Out    ^W Where Is     ^K Cut Text     ^J Justify
^X Exit          ^R Read File    ^\ Replace      ^U Uncut Text   ^T To Spell
```

```
    <!-- Ubuntu OS vulnerabilities -->
    <provider name="canonical">
      <enabled>yes</enabled>
      <os>trusty</os>
      <os>xenial</os>
      <os>bionic</os>
      <os>focal</os>
      <os>jammy</os>
      <update_interval>1h</update_interval>
    </provider>
```

```
    <!-- Debian OS vulnerabilities -->
    <provider name="debian">
      <enabled>yes</enabled>
      <os>buster</os>
      <os>bullseye</os>
      <os>bookworm</os>
      <update_interval>1h</update_interval>
    </provider>
```

Now we have to enable “Aggregate Vulnerabilities” option. When you enable this option the vulnerability database will start downloading after restart wazuh manager.

```
    <!-- Windows OS vulnerabilities -->
    <provider name="msu">
      <enabled>yes</enabled>
      <update_interval>1h</update_interval>
    </provider>
```

```
    <!-- Aggregate vulnerabilities -->
    <provider name="nvd">
      <enabled>yes</enabled>
      <update_interval>1h</update_interval>
    </provider>
```

```
</vulnerability-detector>
```

5

Now restart wazuh manager

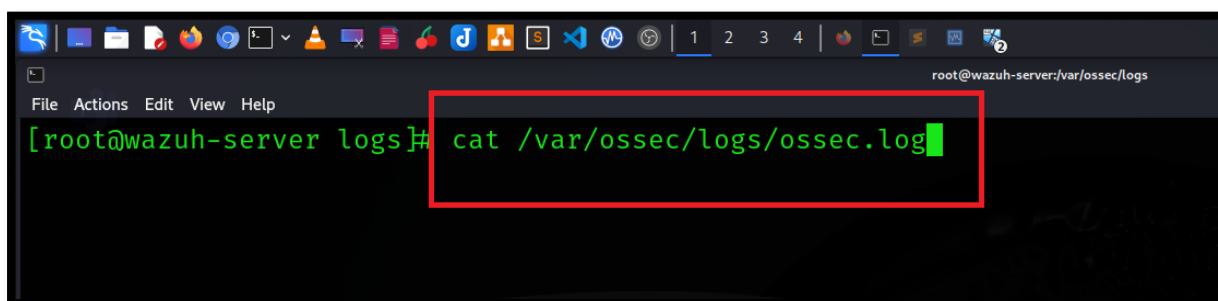
Command: `systemctl restart wazuh-manager`



```
root@wazuh-server~  
[wazuh-user@wazuh-server ~]$ sudo -i  
[root@wazuh-server ~]# nano /var/ossec/etc/ossec.conf  
[root@wazuh-server ~]# systemctl restart wazuh-manager  
[root@wazuh-server ~]#
```

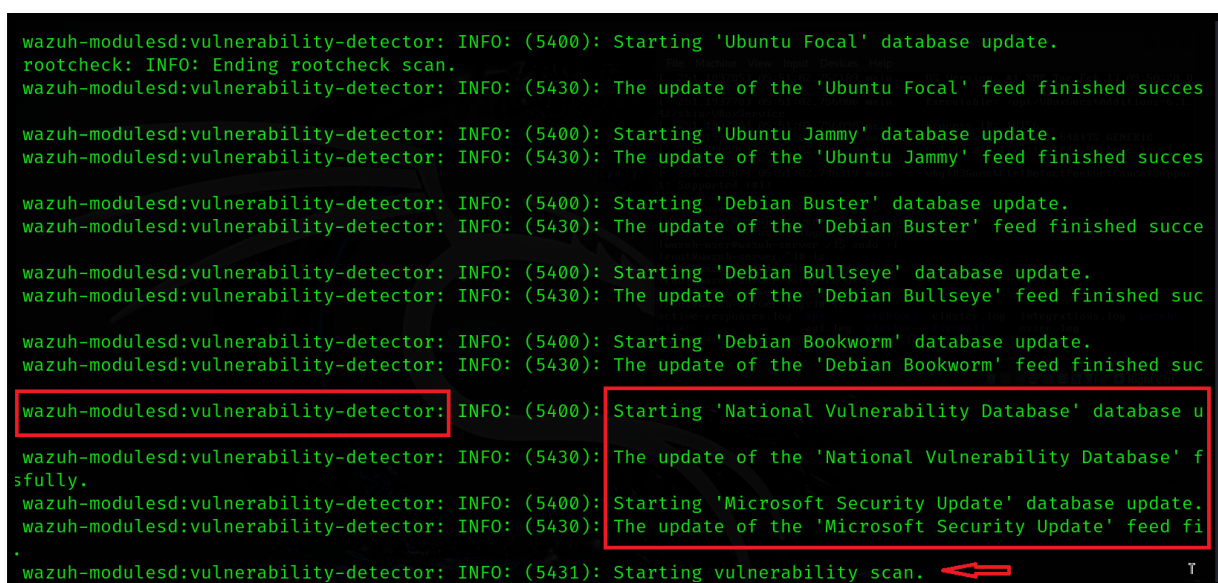
Now check the logs of “ossec.log” file

Command: `cat /var/ossec/logs/ossec.log`



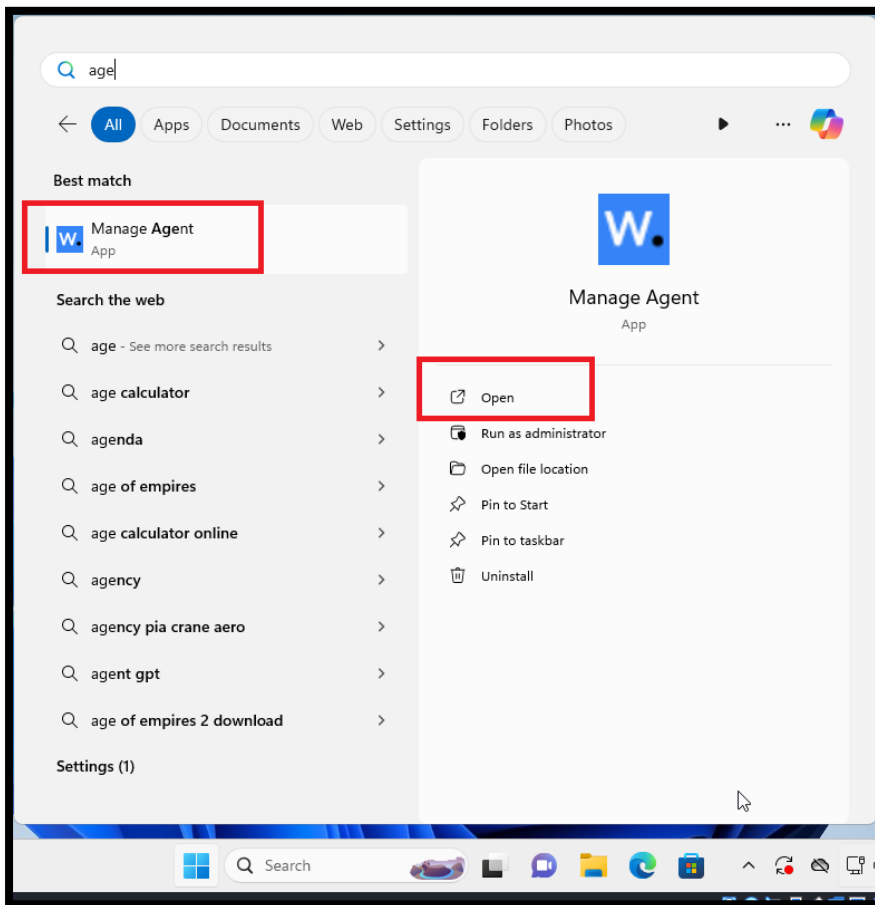
```
root@wazuh-server:/var/ossec/logs  
[root@wazuh-server logs]# cat /var/ossec/logs/ossec.log
```

Vulnerability database is updating and starting scanning. So this vulnerability database is download form: <https://nvd.nist.gov/vuln/data-feeds>

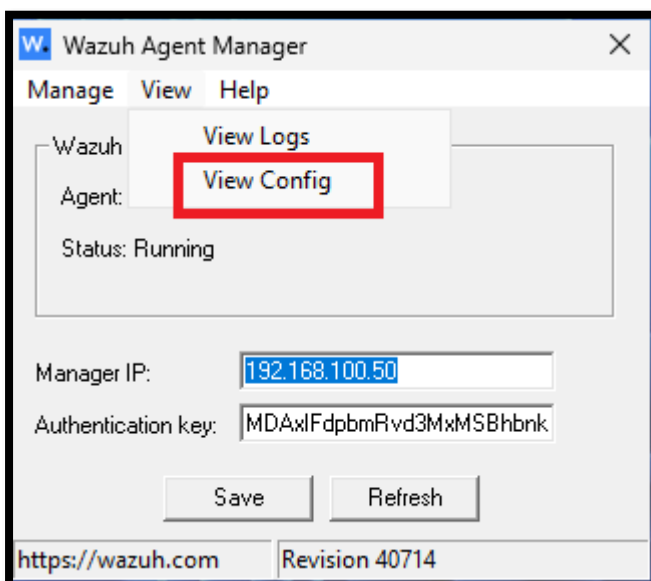


```
wazuh-modulesd:vulnerability-detector: INFO: (5400): Starting 'Ubuntu Focal' database update.  
rootcheck: INFO: Ending rootcheck scan.  
wazuh-modulesd:vulnerability-detector: INFO: (5430): The update of the 'Ubuntu Focal' feed finished succes  
wazuh-modulesd:vulnerability-detector: INFO: (5400): Starting 'Ubuntu Jammy' database update.  
wazuh-modulesd:vulnerability-detector: INFO: (5430): The update of the 'Ubuntu Jammy' feed finished succes  
wazuh-modulesd:vulnerability-detector: INFO: (5400): Starting 'Debian Buster' database update.  
wazuh-modulesd:vulnerability-detector: INFO: (5430): The update of the 'Debian Buster' feed finished succe  
wazuh-modulesd:vulnerability-detector: INFO: (5400): Starting 'Debian Bullseye' database update.  
wazuh-modulesd:vulnerability-detector: INFO: (5430): The update of the 'Debian Bullseye' feed finished suc  
wazuh-modulesd:vulnerability-detector: INFO: (5400): Starting 'Debian Bookworm' database update.  
wazuh-modulesd:vulnerability-detector: INFO: (5430): The update of the 'Debian Bookworm' feed finished suc  
wazuh-modulesd:vulnerability-detector: INFO: (5400): Starting 'National Vulnerability Database' database u  
wazuh-modulesd:vulnerability-detector: INFO: (5430): The update of the 'National Vulnerability Database' f  
sfully.  
wazuh-modulesd:vulnerability-detector: INFO: (5400): Starting 'Microsoft Security Update' database update.  
wazuh-modulesd:vulnerability-detector: INFO: (5430): The update of the 'Microsoft Security Update' feed fi  
wazuh-modulesd:vulnerability-detector: INFO: (5431): Starting vulnerability scan.
```

Now we have to edit configuration in Windows-agent “ossec.conf: file open wazuh-agent.

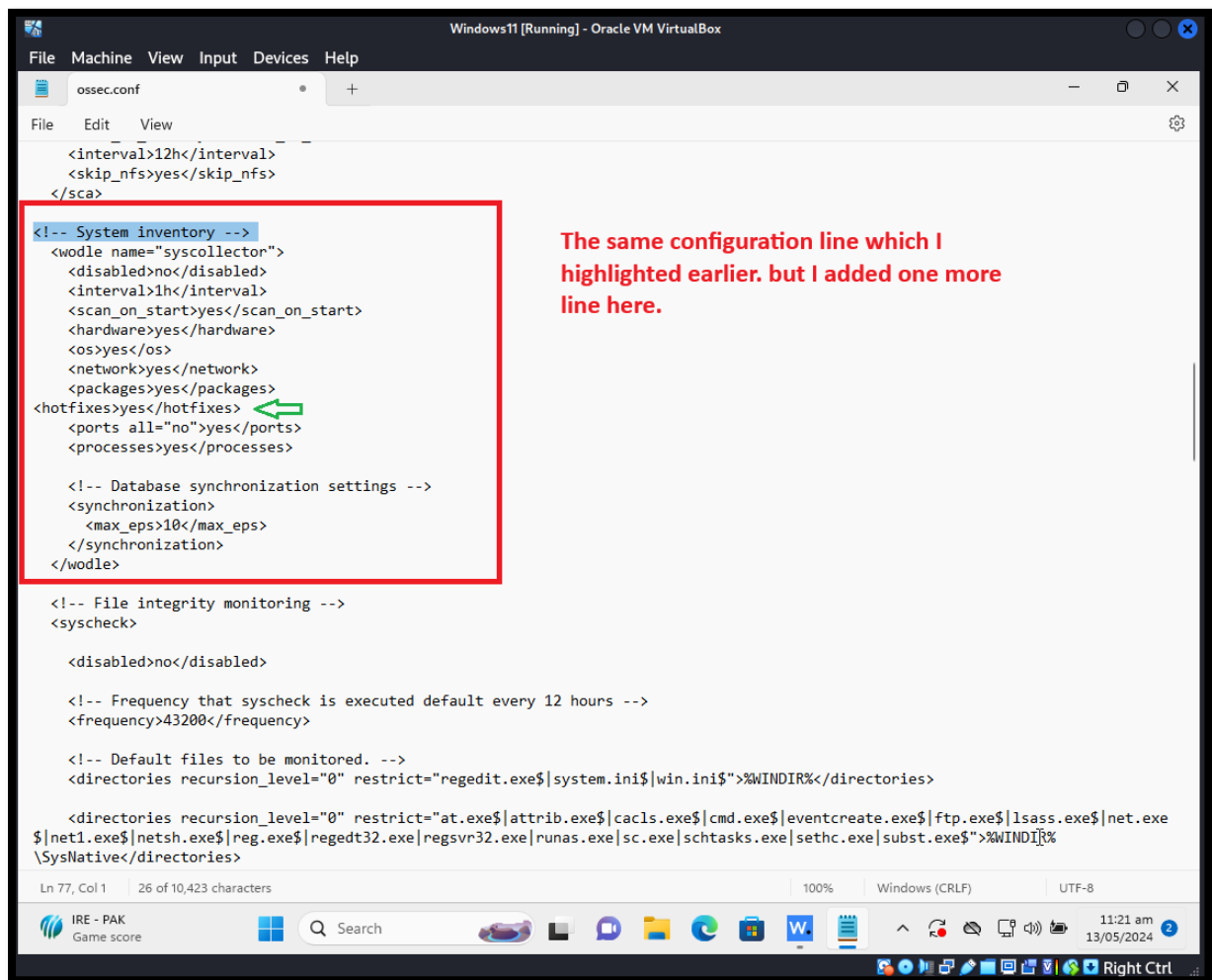


Now go to View > View Config

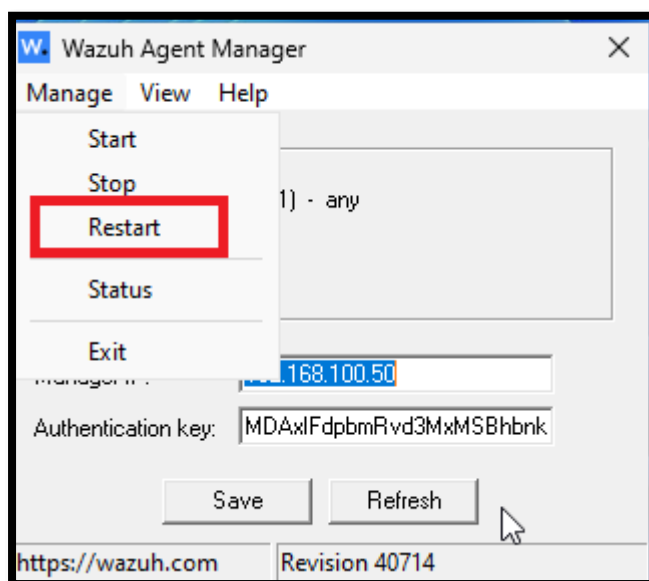


Add the following configuration line here.

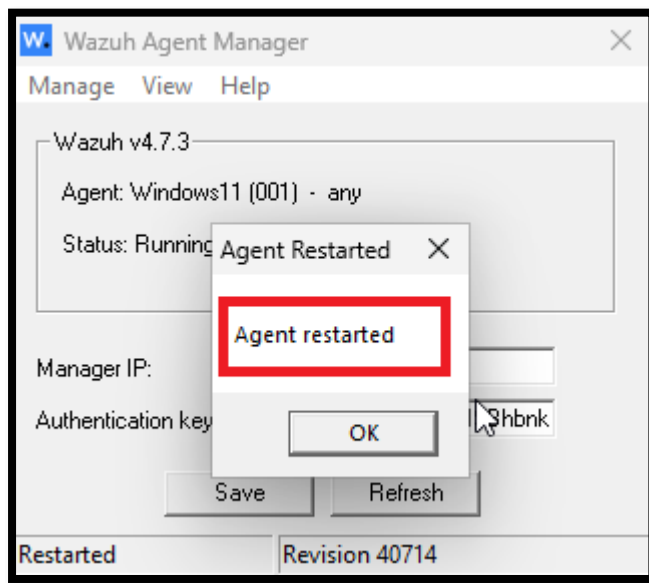
I am adding here one extra line: `<hotfixes>yes</hotfixes>`



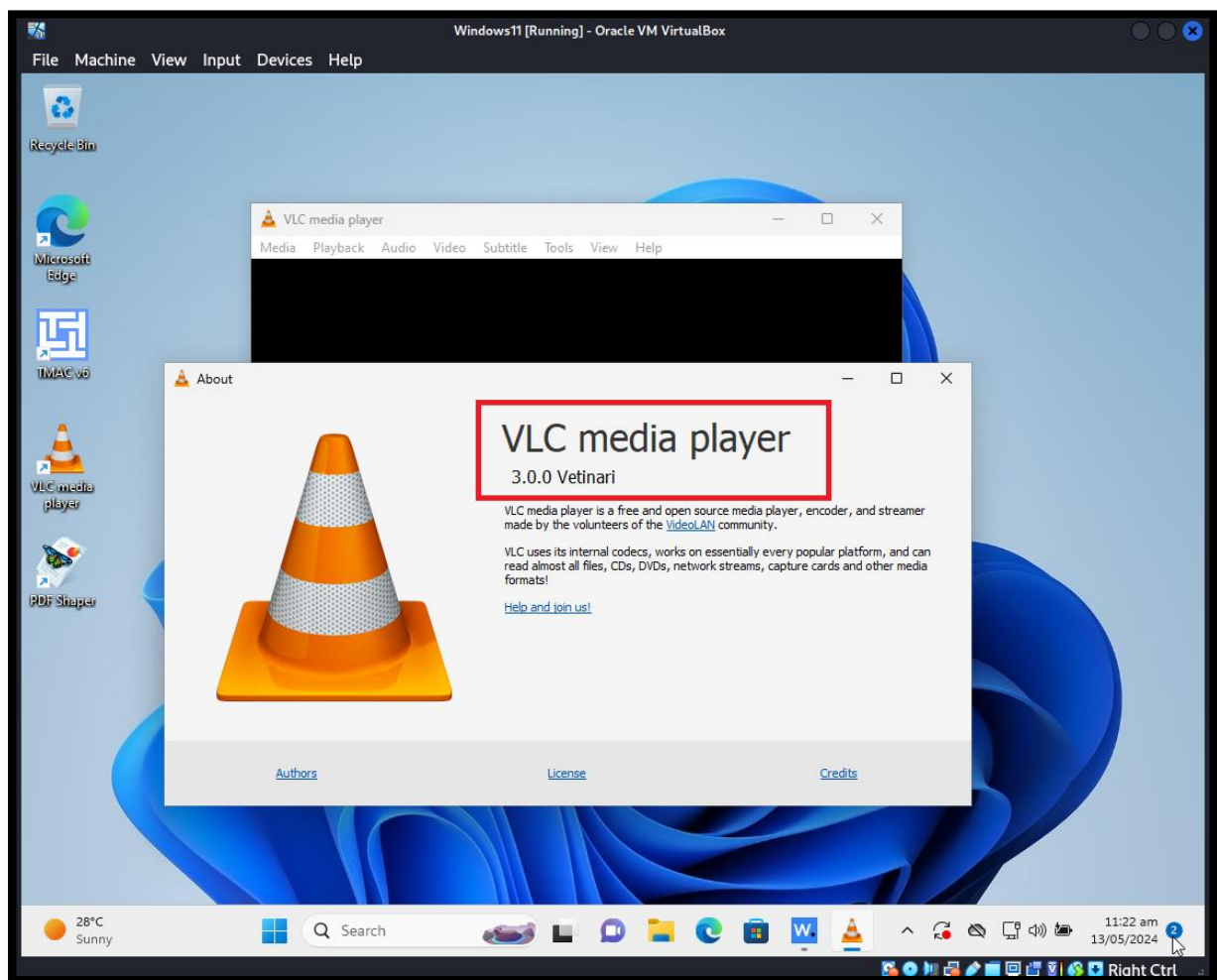
After adding configuration we have to restart wazuh agent.



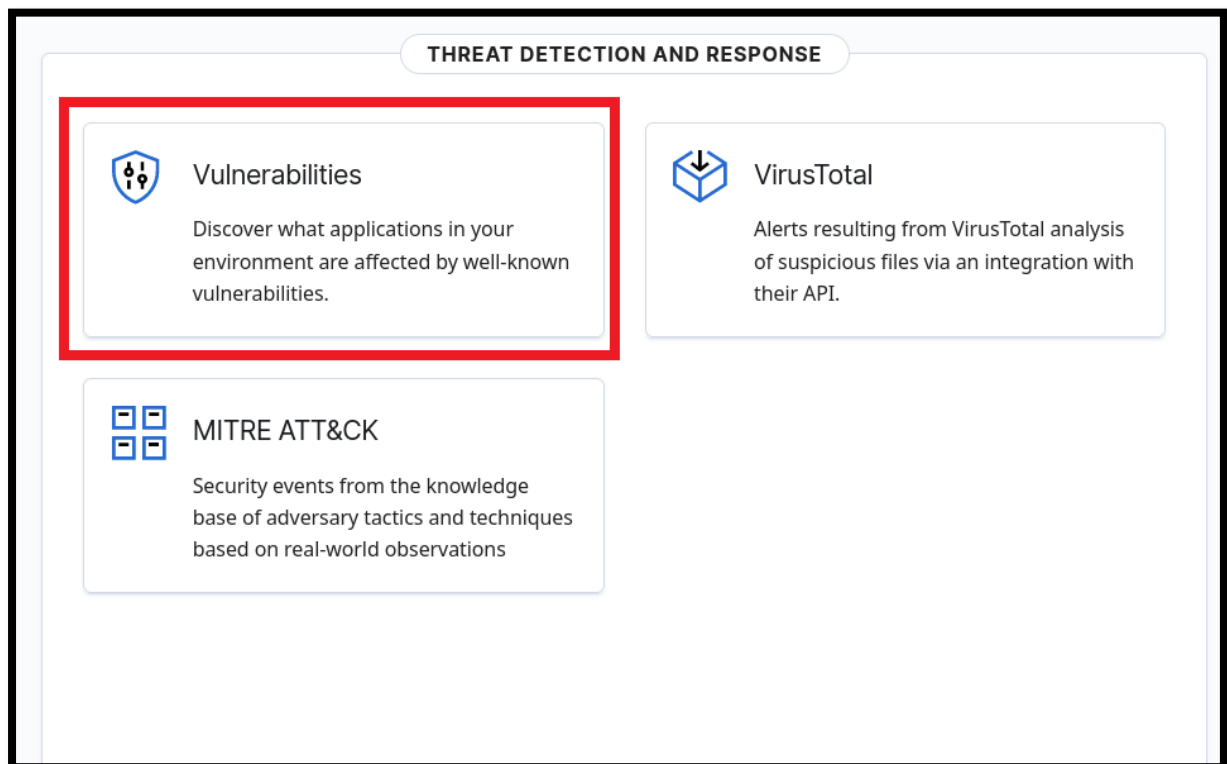
Wazuh-agent is restarted.



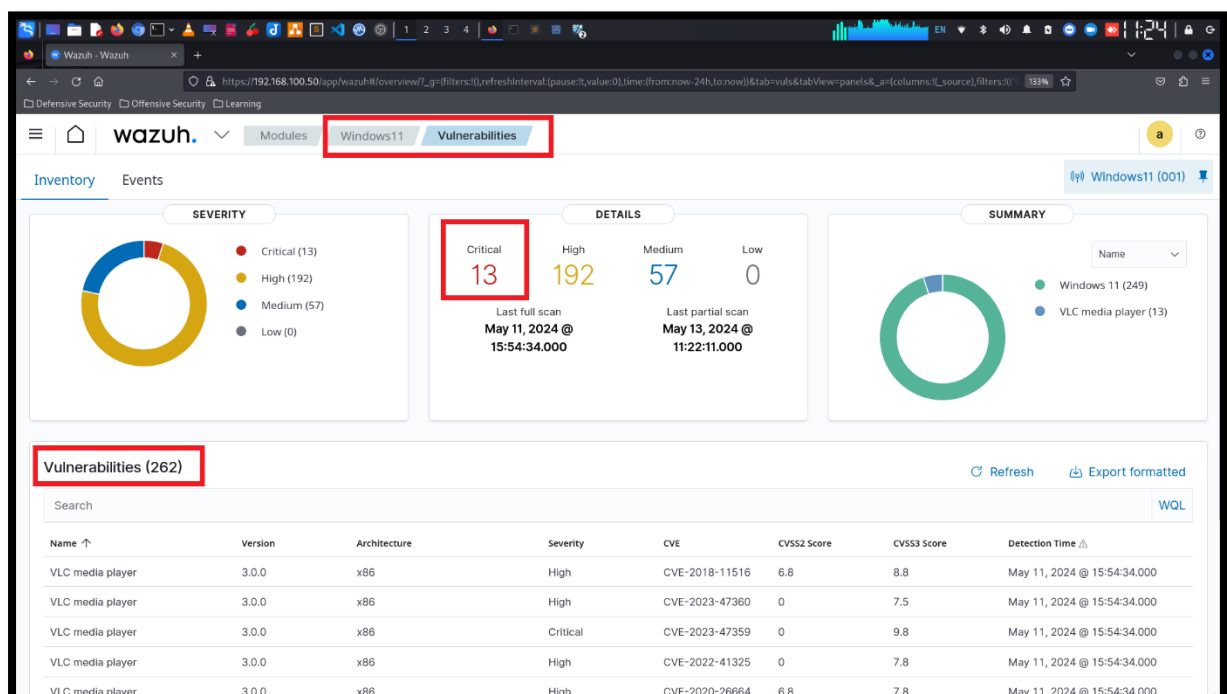
Here is vulnerable VLC media player installed on my Windows 11 machine.



Now click on “Vulnerabilities” again.



Here is **Vulnerabilities Detected!!**



Let's see the vulnerability details. Here is the vulnerability in VLC Media Player.

The screenshot shows the Wazuh Vulnerabilities interface. On the left, a list of vulnerabilities is displayed, all labeled 'VLC media player' with version '3.0.0' and architecture 'x86'. The right panel shows the details for CVE-2018-11516. The 'Title' field is highlighted with a red box, stating 'CVE-2018-11516 affects VLC media player'. Other details include: Name (VLC media player), CVE (CVE-2018-11516), Condition (Version match), Architecture (x86), Version (3.0.0), Last full scan (May 11, 2024 @ 15:54:34.000), Last partial scan (May 13, 2024 @ 11:22:11.000), Published (May 28, 2018 @ 00:00:00.000), Updated (Mar 3, 2023 @ 00:00:00.000), and References (View external references). Below the details, there is a 'Recent events' section with a search bar and a 'Refresh' button. A message at the bottom states 'No results match for this search criteria'.

We have 2 major vulnerabilities in our windows 11.

The screenshot shows the Wazuh Vulnerabilities interface with a list of vulnerabilities. The table has columns: Name, Version, Architecture, Severity, CVE, CVSS2 Score, CVSS3 Score, and Detection Time. The row for 'Windows 11' with CVE-2023-36025 is highlighted with a red box. This vulnerability is High severity, with a CVSS2 Score of 0 and a CVSS3 Score of 8.8. Other vulnerabilities listed include CVE-2019-13602 (High, 6.8 CVSS2, 7.8 CVSS3), CVE-2019-12874 (Critical, 7.5 CVSS2, 9.8 CVSS3), CVE-2019-5439 (Medium, 4.3 CVSS2, 6.5 CVSS3), CVE-2023-38254 (Medium, 0 CVSS2, 6.5 CVSS3), CVE-2023-38186 (Critical, 0 CVSS2, 9.8 CVSS3), CVE-2023-38184 (High, 0 CVSS2, 7.5 CVSS3), CVE-2023-38172 (High, 0 CVSS2, 7.5 CVSS3), CVE-2024-20700 (High, 0 CVSS2, 7.5 CVSS3), and CVE-2024-20699 (Medium, 0 CVSS2, 5.5 CVSS3). The interface includes a search bar, a 'Refresh' button, and an 'Export formatted' button. The bottom of the table shows 'Rows per page: 10' and a pagination bar with page numbers 1 through 27.

Windows 11 Vulnerability details.

The screenshot displays the Wazuh web interface for vulnerability management. On the left, a table lists 262 vulnerabilities, including VLC media player and Windows 11. The main panel shows the details for CVE-2023-36025, which affects Windows 11. The 'Title' field is highlighted with a red box. The details include the version (10.0.22621.1992), architecture (x64), last full scan date (May 11, 2024), last partial scan date (May 13, 2024), and published date (Nov 14, 2023). The condition states that the KB5032190 patch is not installed. The interface also shows a search bar, filters, and a 'Refresh' button.

Name	Version	Arch
VLC media player	3.0.0	x86
VLC media player	3.0.0	x86
VLC media player	3.0.0	x86
Windows 11	10.0.22621.1992	x64
Windows 11	10.0.22621.1992	x64
Windows 11	10.0.22621.1992	x64
Windows 11	10.0.22621.1992	x64
Windows 11	10.0.22621.1992	x64
Windows 11	10.0.22621.1992	x64
Windows 11	10.0.22621.1992	x64

CVE-2023-36025

Details

- Title:** CVE-2023-36025 affects Windows 11
- Name:** Windows 11
- Version:** 10.0.22621.1992
- Architecture:** x64
- Last full scan:** May 11, 2024 @ 15:54:34.000
- Last partial scan:** May 13, 2024 @ 11:22:11.000
- Updated:** Nov 21, 2023 @ 00:00:00.000
- Condition:** KB5032190 patch is not installed
- Published:** Nov 14, 2023 @ 00:00:00.000
- References:** View external references

Recent events 0 hits

Search [] DQL [] This week [] Show dates [] Refresh []

+ Add filter

No results match for this search criteria

SUMMARY

In summary, Wazuh's vulnerability detection capabilities help organizations proactively identify and mitigate security risks, thereby strengthening their defence against cyber threats and ensuring the integrity and availability of their systems and data.